D03 · Yield & Loss Reduction

05 · Standard work & continuous improvement cadence

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1 WHAT IT IS

Standard work is the documented best-known method for performing a repeatable task — sequence of steps, cycle time, and standard WIP — established as the floor on which improvement builds. Continuous improvement cadence is the operating rhythm (daily, weekly, monthly) that sustains standard work, surfaces deviations, and feeds improvement ideas into a structured kaizen pipeline. Together they form the operational discipline that converts Lean from a project portfolio into an operating system; absent either, the system regresses to the mean within 12 months.

2 WHY IT MATTERS

Standard work and CI cadence are the operational disciplines that determine whether a Lean transformation sustains or regresses — and they are the disciplines most consistently abandoned because they are invisible in monthly reports. **The dominant failure mode is documented standard work that is not audited in gemba combined with a kaizen cadence that decays from daily to weekly to quarterly to annual**. The seminal text on the subject quantifies the asymmetry directly **(Imai, 1986 — peer-reviewed, "Kaizen: The Key to Japan's Competitive Success", McGraw-Hill)**: kaizen as an operational discipline (small daily improvements driven by every operator) consistently produces 10–20x more cumulative improvement than kaizen as an event-based program. The economic argument is unambiguous — **plants running daily standard-work audits with active improvement-idea pipelines generate 30–80 implemented ideas per operator per year; plants without that cadence generate 1–3**. The difference is not training, technology, or methodology; it is the operational rhythm that surfaces what to improve and the management discipline that responds to it.

3 HOW IT IS DONE

1. **Document standard work at the level the operator can use, not the level engineering produced.** A 12-page standard operating procedure with technical specifications is a document the operator will not consult during the cycle. The discipline is to convert each procedure into operator-usable standard work: visual sequence (photographs of the steps), cycle-time targets, standard WIP, key quality checks at the point they apply. Engineering's SOP lives in the document repository; the operator's standard work lives at the cell.
2. **Audit standard work in gemba daily — not in management review monthly.** The supervisor walks the cells at the start of each shift, observes 3–5 operators executing the standard work, captures deviation patterns, and either coaches the operator back to standard or surfaces the procedure for revision. The daily gemba walk is the heartbeat of sustained standard work; without it, variation between operators grows by 2–5% per quarter until the standard is fiction.
3. **Treat operator-surfaced deviation as the most valuable signal in the operation — not as a discipline problem.** When an operator does not follow the documented standard, three things may be true: the operator is wrong (coach), the standard is wrong (revise), or the system around the operator is wrong (kaizen). The supervisor's job at the daily gemba walk is to determine which, with bias toward the second and third. The operator who points out that the standard does not work is the most valuable source of improvement ideas in the plant; the supervisor who responds with discipline shuts the source down permanently.
4. **Establish a tiered daily-management cadence — cell standup, line meeting, plant huddle, weekly review.** The cadence is the structural fix that prevents kaizen from decaying into events. Cell standup at shift start (10 minutes, value-stream KPIs, deviation review). Line meeting mid-shift (15 minutes, cross-cell issues, escalations). Plant huddle daily (20 minutes with operations leadership, key escalations, kaizen pipeline status). Weekly review (60 minutes, kaizen completion, idea pipeline metrics). The tiers reinforce each other; missing tiers cause the entire cadence to collapse.
5. **Implement coaching kata — the structured improvement-cycle discipline that makes operator-driven kaizen sustainable.** The kata routine (target condition -> current condition -> obstacles -> next step -> expected outcome) gives operators and supervisors a shared improvement language. Without the kata, "continuous improvement" becomes whatever

the operator thinks is a good idea this week; with it, every improvement attempt is structured as a hypothesis-test cycle that builds organizational learning. The discipline of kata coaching is what converts kaizen from a program into a behavior.

6. **Measure the kaizen pipeline as a leading indicator — ideas submitted, ideas implemented, time-to-implement, savings per idea.** Most Lean programs measure trailing indicators (OEE, FPY, lead time) and miss the leading indicator that predicts whether those trailing numbers will keep improving. The kaizen-pipeline metrics tell whether the improvement engine is running: an operation submitting 10 ideas per operator per year with 80% implementation has an engine; an operation submitting 2 with 30% implementation has a program in regression. The leading metrics give the warning months before the trailing metrics decay.

WORKED EXAMPLE — ILLUSTRATIVE

A CPG manufacturer (1 plant, 6 production lines, 220 operators) emerges from a 24-month Lean transformation with documented standard work for 187 critical operations, kaizen events on the calendar, and OEE = 71% (up from 58%). Three years later: **OEE has drifted back to 64%, operator-to-operator variation in cycle time runs ±35% on the dominant family, kaizen events still happen quarterly but only 11 ideas were submitted plant-wide in the last year, 7 of them implemented**. Diagnosis: standard work is documented but no longer audited in gemba; supervisors moved from daily walks to once-a-week management reviews; the kaizen program calcified into events. The implementation: re-establishes the tiered daily cadence (shift-start cell standups, daily plant huddle, weekly review), trains 14 supervisors in standard-work observation and coaching kata, and installs a simple kaizen-idea board at every cell with weekly idea review. Results at 12 months: **operator cycle-time variation drops from ±35% to ±8%, FPY rises 6.4 points, OEE recovers from 64% to 76%, kaizen ideas submitted go from 11/year to 340/year with 84% implementation rate**. The standard work documentation did not change; the audit cadence did. The improvement engine that had stopped running is running again — because the daily rhythm exists again.

IN PRACTICE

Audit standard work in gemba daily, not in management review monthly

The standard work that lives in the document repository is theoretical; the one operators actually do is the operational state. A daily gemba walk with the supervisor at the cell is the only audit that surfaces the gap — and the gap is where the improvement lives.

Tiered daily-management cadence as non-negotiable

Plant manager's calendar shows the daily huddle; operations director's calendar shows the weekly review; supervisors are evaluated on coaching cadence, not firefighting velocity. Without leadership commitment that the cadence is non-negotiable, it is the first casualty of the next production crisis.

Operator-surfaced ideas as the leading indicator

Plants with active improvement-idea pipelines generate 30–80 implemented ideas per operator per year; plants without that cadence generate 1–3 **(Imai, 1986 — peer-reviewed, McGraw-Hill)**. Pipeline volume is the leading indicator of whether the cadence is real or ceremonial.

Coaching kata as the structured improvement language

The Toyota distinction: kaizen events solve discrete problems; the improvement kata builds the capability that solves the next thousand problems autonomously. Supervisors are coaches, not firefighters — and the kata is the script that makes coaching repeatable.

WHAT YOU WOULD USE

STANDARD WORK & CI CADENCE

Mevisio / Tervene / KaiNexus — Digital daily-management platforms with tiered standup boards, KPI visualizations, idea pipelines, and coaching workflow — the right fit for multi-cell, multi-line operations standardizing the cadence across supervisors **(Mevisio — vendor; Tervene — vendor; KaiNexus — vendor)**.

SafetyCulture (iAuditor) / Tulip for standard-work audit — Mobile-first platforms for supervisor gemba walks and standard-work observation, with photo capture, deviation tracking, and trend analytics; strong fit for the audit-cadence piece specifically **(SafetyCulture — vendor; Tulip — vendor)**.

Toyota Kata coaching providers + internal master trainer — External providers (LEI, Kata School Cascadia, regional consultants) for foundational kata training, paired with internal master trainer to sustain the practice; the right structure when the operational behavior change is the central objective **(Lean Enterprise Institute — vendor)**.

Physical cell boards + supervisor walk log + monthly portfolio review — The original Toyota approach — laminated standard-work sheets at the cell, paper kaizen-idea boards with weekly review, supervisor walk log in a notebook. The lowest-tech option produces the deepest cultural anchor when the cadence is real; the artifacts are digitized monthly for trend analytics but the daily work happens with paper.

4 SO WHAT

Standard work that is documented but not audited in gemba is a procedure that operators ignore; continuous improvement without a daily cadence is a quarterly panic dressed up as a program. Document standard work at the operator's level not engineering's, audit in gemba daily not in management review monthly, treat operator-surfaced deviation as the most valuable signal in the operation, establish a tiered daily-management cadence that cannot be cut under production pressure, implement coaching kata as the structured improvement language, and measure the kaizen pipeline as a leading indicator. The transformation projects make the gain; the cadence is what keeps it. The plants that sustain Lean over decades win not because their methodology is better but because their rhythm is real.

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